

Gene Pulser® Electroprotocol

Cell Type Plant

Molecules Electroporated DNA: plasmid pTZ 18U with a 3 kB insert (total: 6 kB), circular.

Species Used *Nicotiana plumbaginifolia*; protoplasts from leaf

Before the Pulse

Cell Growth Medium 5 mM CaCl₂; 0.5 M sucrose, 0.3% Macerozyme R-10, 0.3% cellulose "Onozuka" R-10, pH 4.7 (enzymes-Yokult Honsha Co.,Ltd.)

Growth Phase at Harvest Not given

Pre-pulse Incubation 2 hours at 4°C

Wash Solution 4 mM CaCl₂, 80mM KCl, 8.0% Mannitol, 2 mM Na₂P0₄, pH 7.2

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature 2 hr. at 4 °C

Electroporation Medium Same as wash solution

Cuvette Gap 0.2 cm

Cell Density 10 (6) / ml

Voltage 0.16 kV

Volume of Cells 300 µl

Field Strength 0.8 kV/cm

DNA Concentration 3 µg / µl

DNA Resuspension Buffer TE Buffer (10 mM Tris, 1 mM EDTA, pH 8.0)

Capacitor 125 µF

Volume of DNA 10 µg DNA / pulse, 3 µl.

Resistor (Pulse Controller) none Ω

After the Pulse

Time Constant 2.8 msec

Outgrowth Medium 9% Mannitol, 3% sucrose, 100 µg/ml cefotamine (antibiotic)

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature 25 °C

Length of Incubation 36 hours

I am no longer a member of this laboratory, but work is in progress there.

Selection Method or Assay Used Transient expression assays

Electroporation Efficiency Not determined

Per Cent Survival Not determined

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Survey Number

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